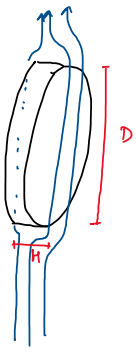
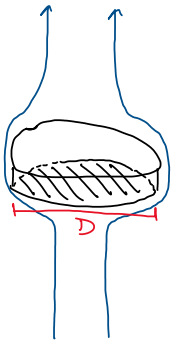
Drag coefficient

Monday, 28 August 2023 10:11 AM



Characteristic Area: $H \cdot D$

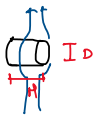


Characteristic Area: $\pi \frac{D^2}{4}$

If the disk was very small:
w/ a much large H
(i.e. a cylinder)

↓

Then characteristic area of circumference would make the most sense



Characteristic area: $\pi D \cdot H$

Either way, providing a C_D w/ reference area taken for calculated is the convention

Week 3

Monday, September 4, 2023 4:08 PM

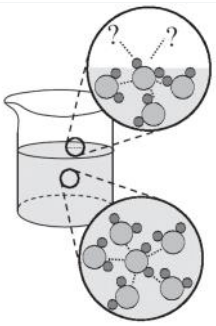
Recitation 09/05 draft

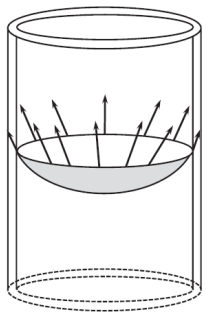
Monday, September 4, 2023 3:47 PM

Concept Review

Surface Tension

- Now we are dealing with interactions b/w two fluid at their interface (e.g. bubbles)
- Phase boundary = "free surface" of the flow
- Surface tension is a fluid property (like density and viscosity; denoted by σ $[\frac{N}{m}]$ and is a result of unbalanced molecular forces at a surface
- Force is tangent to the surface and normal to the boundary

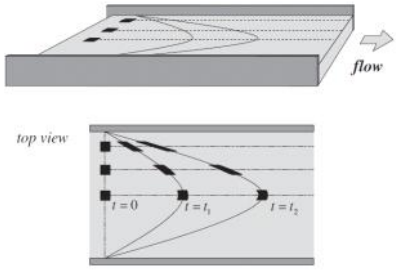




the downward force on surface particles is not balanced by an upward force of fluid on the other side of the interface, the net effect is that the *free surface behaves like a thin massless film under tension.*

Reynolds Transport Theorem

- We could apply balance laws on each fluid element (box) --> but have to deal with awkward shapes
- We can define a control volume - an arbitrary shape region that is fixed within the flow and sees fluid coming and going out



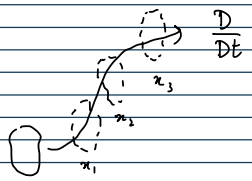
Extensive property : Depends on the amount of matter in a sample Ψ

Intensive property : Independent of the size of sample $\psi = \frac{\partial \Psi}{\partial m}$ ← mass

Total material Volume corresponding to a mass w/ the property Ψ is written as: $\Psi = \iiint_{V(t)} \rho \cdot \psi \, dV$ ← arbitrary material volume

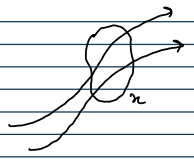
Frames of Reference :

Material Volume vs. Control volume



+ Flux of a property across a material volume is 0
  - i.e. Volume whose boundaries are moving w/ time

+ Rate of change of a fluid property given by: $\left[\frac{D}{Dt}\right]$



+ Fixed x-position; there is a non-zero flux

+ Rate of change of a fluid property given by: $\left[\frac{\partial}{\partial t}\right]$

$$\boxed{\frac{D}{Dt}[FP] = \frac{\partial}{\partial t}[FP] + v \cdot \nabla(FP)}$$

FP = Fluid property
v = velocity of fluid

We know: $\Psi = \iiint_{V(t)} \rho \Psi \, dV$ → This is w/ respect to material frame of reference

- However, the volume is changing so we cannot evaluate (as we don't know limits of integration)
- But we know we can go from material to control frame of reference via the green equation

$$\frac{D\Psi}{Dt} = \frac{\partial}{\partial t}\Psi + v \cdot \nabla\Psi$$

$$= \frac{\partial}{\partial t}\left[\iiint_{V_0}\right.$$

Momentum Balance

Momentum (p) = Mass (m) · Velocity (V)

$$\underline{p} = m \cdot \underline{v}$$

Mass = (Density) · Volume

$$(m) = \rho \cdot \int V$$

$$\underline{p} = (\rho \cdot V) \cdot \underline{v}$$

m^3 term

$m^2 \cdot m$

Rate of Momentum Change: Momentum per unit time

"dividing by time"

$$\underline{p} = (\rho \cdot \text{Area} \cdot l) \cdot \underline{v}$$

$$\sim \frac{\underline{p}}{\Delta t} = \rho \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot v$$

($\frac{l}{\Delta t} \sim v$)

That is why momentum balance has two velocity terms!

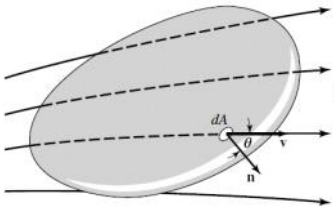
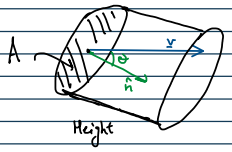


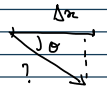
Figure 5.2 Fluid flow through a control volume.

We care about fluid velocity in the direction of the normal to the surface



→ Volume of fluid passed through cross-section A:

$V_{cylinder} = A \cdot \text{Height}$ ← unknown



$\frac{?}{\Delta x} = \cos\theta$

$? = \Delta x \cos\theta$

$\therefore V_{cylinder} = A \cdot \Delta x \cdot \cos\theta$

$\Delta V = A \cdot \Delta x \cos\theta$

$\frac{\Delta V}{\Delta t} = A \cdot \frac{\Delta x}{\Delta t} \cos\theta = A \cdot \underline{v} \cdot \cos\theta$

$\frac{\Delta x}{\Delta t} = \underline{v}$

$\hat{n} \cdot \underline{v} = |\hat{n}|\,|\underline{v}| \cos\theta = v\cos\theta$ (1)

Volumetric flow rate through A : $Q = v \cdot \cos\theta \; A = (\hat{n} \cdot \underline{v}) A$

↓

Mass flow = Density · Volumetric flow rate

$m = \rho \cdot Q$

$m = \rho \cdot (\hat{n} \cdot \underline{v}) A$

Momentum flow of liquid through A $\Rightarrow \underline{P} = m \cdot \underline{v}$

$= \rho (\hat{n} \cdot \underline{v}) A \, \underline{v}$

↳ Now we chose arbitrary area → Taking a differential area & integrating momentum quantity over the entire control surface

Net momentum change from CV : $\iint_S \underline{v}\, \rho (\hat{n} \cdot \underline{v})\, dA$

$\Rightarrow$ (In − out) = change

Recitation 09/05 Master

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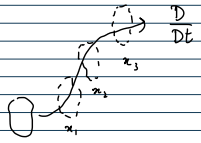
Concept Review

Reynolds Transport Theorem

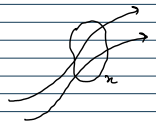
- Has different forms depending on the frame of reference we are in

Frames of Reference:

Material Volume vs. Control volume



+ Flux of a property across a material volume is 0
  - i.e. Volume whose boundaries are moving w/ time
+ Rate of change of a fluid property given by: $\left[\frac{D}{Dt}\right]$



+ Fixed x-position; there is a non-zero flux
+ Rate of change of a fluid property given by: $\left[\frac{\partial}{\partial t}\right]$

$$\frac{D}{Dt}[FP] = \frac{\partial}{\partial t}[FP] + v \cdot \nabla(FP)$$

FP = Fluid property
v = velocity of fluid

Material Coordinates

$$\frac{D\Psi}{Dt} = \int_{V_a(t)} \frac{\partial \psi}{\partial t} \rho dV + \int_{A_a(t)} \psi \rho \underline{v} \cdot \hat{n} dA_a$$

↳ General form (Can be applied to a particular fluid property such as mass, energy, momentum)

Control Volume coordinates

Reynolds transport theorem (momentum balance on CV)

$$\sum_{\text{on CV}} \underline{f} = \frac{d\mathbf{P}}{dt} + \iint_{\text{CS}} (\hat{n} \cdot \underline{v}) \rho \underline{v} \, dS \qquad (3.137)$$

Momentum Balance

Momentum (p) = Mass (m) · Velocity (V)

$\underline{p} = m \cdot \underline{v}$

Mass = (Density) · Volume

$m = \rho \cdot \forall$

$\underline{p} = (\rho \cdot \forall) \cdot \underline{v}$

m^3 term

$m^2 \cdot m$

Rate of Momentum Change: Momentum per unit time

"dividing by time"

$\underline{p} = (\rho \cdot \text{Area} \cdot l) \cdot \underline{v}$

$\approx \underline{p} = \rho \cdot \text{Area} \cdot \frac{l}{t} \cdot v$

"dividing by time" ↳ $\underline{F} = (p \cdot \text{Area} \cdot l) \cdot \underline{v}$

$\approx \frac{\underline{F}}{\Delta t} = p \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot \underline{v}$ ($\frac{l}{\Delta t} = v$)

That is why momentum balance has two velocity terms!

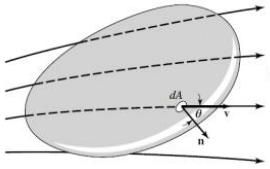
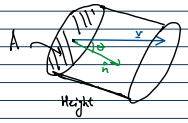


Figure 5.2 Fluid flow through a control volume.

We care about fluid velocity in the direction of the normal to the surface



→ Volume of fluid passed through cross-section A:

$V_{cylinder} = A \cdot \text{Height}$ (unknown)

$\frac{?}{\Delta x} = \cos\theta$

$? = \Delta x \cos\theta$

$\therefore V_{cylinder} = A \cdot \Delta x \cdot \cos\theta$

$\Delta V = A \cdot \Delta x \cos\theta$

$\frac{\Delta V}{\Delta t} = A \cdot \frac{\Delta x}{\Delta t} \cos\theta = A \cdot \underline{v} \cdot \cos\theta$ ($\frac{\Delta x}{\Delta t} = \underline{v}$)

$\hat{n} \cdot \underline{v} = |\hat{n}|\,|\underline{v}| \cos\theta = v\cos\theta$ ($|\hat{n}| = 1$)

Volumetric flow rate through A: $Q = v \cdot \cos\theta \, A = (\hat{n} \cdot \underline{v}) A$

Mass flow = Density · Volumetric flow rate

$m = \rho \cdot Q$

$m = \rho \cdot (\hat{n} \cdot \underline{v}) A$

Momentum flow of liquid through A $\Rightarrow \underline{P} = m \cdot \underline{v}$

$= \rho (\hat{n} \cdot \underline{v}) A \, \underline{v}$

↳ Now we choose arbitrary area → Taking a differential area & integrating momentum quantity over the entire control surface

Net momentum change from CV : $\iint_S \underline{v}\, \rho (\hat{n} \cdot \underline{v})\, dA$

$\Rightarrow$ (In − out) = change

Problems

PROBLEM: 3.15 What is the flow of momentum through the hemispherical surface described in Problem 14?

PROBLEM: 3.14 A uniform flow $\underline{v} = U\hat{e}_z$ of an incompressible fluid of density ρ passes through a volume that is in the shape of a half sphere of radius R. The outwardly pointing unit normal of the flat surface of the half sphere is $\hat{n} = -\hat{e}_z$. What is the mass flow rate through the hemispherical surface of this volume? Show that you can obtain the correct answer by integrating the formal expression for Q (Equation 3.87).

PROBLEM: 3.21 The y-component of a velocity field in flow through a slit (Cartesian coordinates) is given here. What is the average value of the velocity? $2H$ is the gap between the plates. At what location is the velocity a maximum? The units on velocity and A are m/s and the units of x and H are m.

$$\underline{v} = \begin{pmatrix} 0 \\ v_y(x) \\ 0 \end{pmatrix}_{xyz}$$

$$v_y(x) = A\left(1 - \frac{(x-H)^2}{H^2}\right)$$

PROBLEM: 3.31 For the Cartesian coordinate system shown in Figure 3.38, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?

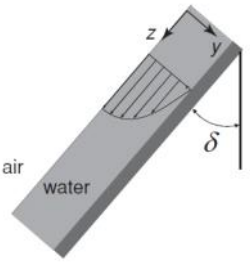


Figure 3.38 Flow coordinate system for Problem 31.



Figure 3.38 Flow coordinate system for Problem 31.

Recitation 09/05

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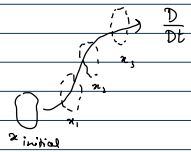
Concept Review

Reynolds Transport Theorem

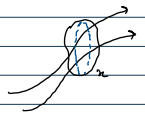
- Has different forms depending on the frame of reference we are in

Frames of Reference:

Material Volume vs. Control volume



+ Flux of a property across a material volume is 0
  ↳ i.e. Volume whose boundaries are moving w/ time
+ Rate of change of a fluid property given by: $\left[\frac{D}{Dt}\right]$



+ Fixed x-position; there is a non-zero flux
+ Rate of change of a fluid property given by: $\left[\frac{\partial}{\partial t}\right]$

$$\frac{D}{Dt}[FP] = \frac{\partial}{\partial t}[FP] + v \cdot \nabla(FP)$$

FP = Fluid property
v = velocity of fluid



Extensive var: $\Psi \neq \psi$ (Intensive var: Independent of mass)

$\Psi = \frac{\partial \psi}{\partial m}$

Material Coordinates

$$\frac{D\Psi}{Dt} = \int_{V_a(t)} \frac{\partial \psi}{\partial t} \rho dV + \int_{A_a(t)} \psi \rho \underline{v} \cdot \hat{n} dA_a$$

↳ General form (Can be applied to a particular fluid property such as mass, energy, momentum)

Reynolds transport theorem (momentum balance on CV)

Control Volume coordinates

$$\sum_{\text{on CV}} \underline{f} = \frac{d\mathbf{P}}{dt} + \iint_{CS} (\hat{n} \cdot \underline{v}) \rho \underline{v} \, dS \quad (3.137)$$

(momentum; force; Control surface)

Momentum Balance

Momentum (p) = Mass (m) · velocity (V)

$p = m \cdot \underline{v}$

Mass = (Density) · Volume

$m = \rho \cdot V$

$p = (\rho \cdot V) \cdot \underline{v}$ (Density, volume, velocity)

m^3 term

$m^2 \cdot m \rightarrow$ Area · length

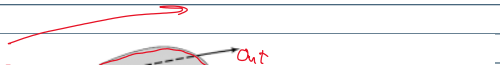
Rate of Momentum Change: Momentum per unit time

"dividing by time"

$\underline{f} = (\rho \cdot \text{Area} \cdot l) \cdot \underline{v}$

$\approx \frac{\underline{f}}{\Delta t} = \rho \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot v$ ($\frac{l}{\Delta t} = v$)

That is why momentum balance has two velocity terms!



That is why momentum balance has two velocity terms!

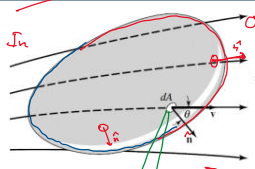
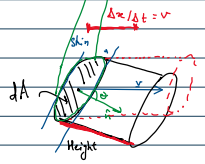


Figure 5.2 Fluid flow through a control volume.

In − out (eger/−cosθ) = 0

We care about fluid velocity in the direction of the normal to the surface



→ Volume of fluid passed through cross-section A:

$V_{cylinder} = A \cdot \text{Height}$ (unknown)

SOH CAH TOA



$\frac{?}{\Delta x} = \cos\theta$

$? = \Delta x \cos\theta$

$\therefore V_{cylinder} = A \cdot \Delta x \cos\theta$ → Change in Volume of cylinder

$\Delta V = A \cdot \Delta x \cos\theta$

$\frac{\Delta V}{\Delta t} = A \cdot \frac{\Delta x}{\Delta t} \cos\theta = A \cdot \underline{v} \cdot \cos\theta$ ($\frac{\Delta x}{\Delta t} = \underline{v}$)

$V = \bar{V}$

$\underline{v} = \bar{v} = v$

$a \cdot b = |a||b| \cos\theta$

$\hat{n} \cdot \underline{v} = |\hat{n}||\underline{v}| \cos\theta = v\cos\theta$

Volumetric flow rate through A: $\frac{\Delta V}{\Delta t} = Q = v \cdot \cos\theta \, A = \boxed{(\hat{n} \cdot \underline{v}) A}$

Mass flow = Density · Volumetric flow rate

$m = \rho \cdot Q$

$\boxed{m = \rho \cdot (\hat{n} \cdot \underline{v}) dA}$

Momentum flow of liquid through A $\Rightarrow$ $d\underline{p} = m\underline{v}$

$\boxed{d\underline{p} = \rho (\hat{n} \cdot \underline{v}) A \, \underline{v}}$

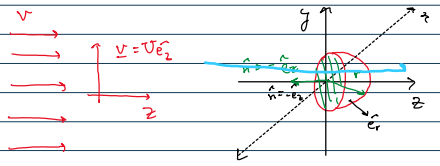
→ Now we chose arbitrary area → Taking a differential area & integrating momentum quantity over the entire control surface

Net momentum change from CV: $\iint_S \underline{v} \rho (\hat{n} \cdot \underline{v}) \, dA$ ⇒ (In − out) = change

Efflux [In − out] | Surface | Area

Problems

PROBLEM: 3.15 What is the flow of momentum through the hemispherical surface described in Problem 14?



Momentum change / flow rate $= \iint_A \rho \underline{v} (\hat{n} \cdot \underline{v}) \, dS$

ρ: uniform, constant

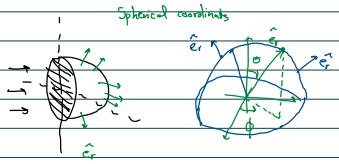
$\frac{d\underline{p}}{dt} = \rho \underline{v} \iint_A (\hat{n} \cdot \underline{v}) \, dS$

PROBLEM: 3.14 A uniform flow $\underline{v} = V\hat{e}_z$ of an incompressible fluid of density ρ passes through a volume that is in the shape of a half sphere of radius R. The outwardly pointing unit normal of the flat surface of the half sphere is $\hat{n} = -\hat{e}_z$. What is the mass flow rate through the hemispherical surface of this volume? Show that you can obtain the correct answer by integrating the formal expression for Q (Equation 3.87).

$\hat{e}_z$ | Accumulation = 0

Mass balance → Q = 0

Spherical coordinates



$(\hat{n} \cdot \underline{v})$

$\hat{n} = \hat{e}_r$ | $V\hat{e}_z$

$\therefore (\hat{e}_r \cdot \underline{v})$

$(\hat{e}_r \cdot V\hat{e}_z)$

(Unit basis commonly representation of

$\hat{n} = \hat{e}_r$, $v\hat{e}_z$

$\therefore (\hat{e}_r \cdot \underline{v})$

$(\hat{e}_r \cdot v\hat{e}_z)$

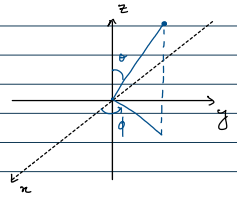
(Unit basis conversion) expression of spherical in cartesian

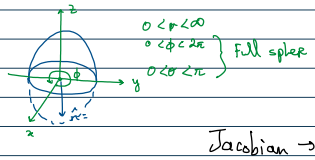
$\hat{e}_r = \sin\theta\cos\phi\, e_x + \sin\theta\sin\phi\, \hat{e}_y + \cos\theta\, \hat{e}_z$

$\begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}_{xyz} \cdot \begin{pmatrix} 0 \\ 0 \\ v \end{pmatrix}_{xyz} = v\cos\theta$

$\frac{dp}{dt} = \rho \underline{v} \iint_A (\hat{n}\cdot\underline{v})\, dS$, $v\cos\theta$

$= \rho \underline{v} \iint_A v\cos\theta\, dS$; $\underline{v}$ constant





$0 < r < \infty$
$0 < \phi < 2\pi$ } Full sphere
$0 < \theta < \pi$

Jacobian → $R^2 \sin\theta\, d\theta\, d\phi$

momentum ↓ density ↓

$\frac{\Delta p}{\Delta t} = \frac{dp}{dt} = \rho v \int_0^{2\pi}\int_0^{\pi/2} (\hat{e}_r \cdot \underline{v})\, d\theta\, d\phi$, $v\cos\theta$

$= \rho v \int_0^{2\pi}\int_0^{\pi/2} v\cos\theta\, (R^2\sin\theta)\, d\theta\, d\phi$ } Solve this !

PROBLEM: 3.21 The y-component of a velocity field in flow through a slit (Cartesian coordinates) is given here. What is the average value of the velocity? $2H$ is the gap between the plates. At what location is the velocity a maximum? The units on velocity and A are m/s and the units of x and H are m.

$$\underline{v} = \begin{pmatrix} 0 \\ v_y(x) \\ 0 \end{pmatrix}_{xyz} \quad \begin{pmatrix} v_x(x,y,z) \\ 0 \\ 0 \end{pmatrix}$$

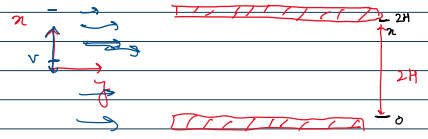
$v_y(x)$ dependence

$$v_y(x) = A\left(1 - \frac{(x-H)^2}{H^2}\right)$$

flow

$$\text{Average of } f(z) = \langle f \rangle = \frac{\int_{z_{min}}^{z_{max}} f(z)\, dz}{(z_{max} - z_{min})} \qquad \frac{\text{Total}}{\#} \qquad (3.41)$$

⟨Ashu⟩



$\langle v_y \rangle = \frac{\int_0^{2H} v_y(x)\cdot dx}{2H - 0} = \frac{1}{2H}\int_0^{2H} A\left(1 - \frac{(x-H)^2}{H^2}\right) dx$

$\langle v_y \rangle = \frac{A}{2H}\int_0^{2H} \left(1 - \frac{(x-H)^2}{H^2}\right) dx$

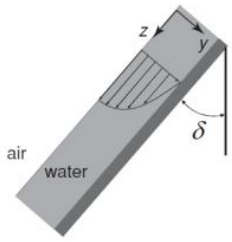
$$\boxed{\langle v_y \rangle = \frac{2}{3}A}$$

$\frac{d v_y}{dx} = \frac{d}{dx}\left(A\left(1 - \frac{(x-H)^2}{H^2}\right)\right) = 0$

$x = H$, $\frac{d v_y}{dx} = 0$

$$\boxed{v_y(x=H) = A}$$

PROBLEM: 3.31 For the Cartesian coordinate system shown in Figure 3.38, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?



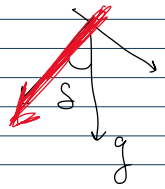


Figure 3.38 Flow coordinate system for Problem 31.

Recitation 09/07

Monday, September 4, 2023 3:47 PM

Concept Review

Reynolds Transport Theorem

- Has different forms depending on the frame of reference we are in

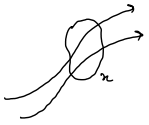
Frames of Reference:

Material Volume → Experiments

vs.

Control volume





+ Flux of a property across a material volume is 0
  - i.e. Volume whose boundaries are moving w/ time
+ Rate of change of a fluid property given by: $\left[\frac{D}{Dt}\right]$

+ Fixed x-position; there is a non-zero flux
+ Rate of change of a fluid property given by: $\left[\frac{\partial}{\partial t}\right]$

$$\frac{D}{Dt}[FP] = \frac{\partial}{\partial t}[FP] + v \cdot \nabla(FP)$$

FP = Fluid property
v = velocity of fluid

$\Psi \neq \psi$

- Ψ: Extensive
- ψ: Intensive → Independent of mass or size

Material Coordinates

Ψ →

$$\frac{D\Psi}{Dt} = \int_{V_a(t)} \frac{\partial \psi}{\partial t} \rho dV + \int_{A_a(t)} \psi \rho \underline{v} \cdot \hat{n} dA_a$$

↳ General form (Can be applied to a particular fluid property such as mass, energy, momentum)

via Newton's second law

Control Volume coordinates

Reynolds transport theorem (momentum balance on CV)

$$\sum_{\text{on CV}} \underline{f} = \frac{d\mathbf{P}}{dt} + \iint_{CS} (\hat{n} \cdot \underline{v}) \rho \underline{v} \, dS \qquad (3.137)$$

- $\sum \underline{f}$: forces on CV
- $\frac{d\mathbf{P}}{dt}$: momentum
- $\iint_{CS} (\hat{n} \cdot \underline{v}) \rho \underline{v} \, dS$: In - out

Momentum Balance

Momentum (p) = Mass (m) · velocity ($\underline{v}$)

$\underline{p} = m \cdot \underline{v}$

Mass (m) = (Density) · Volume = $\rho \cdot V$

$\underline{p} = (\rho \cdot V) \cdot \underline{v}$

m^3 term

$m^2 \cdot m$

Rate of Momentum Change: Momentum per unit time

"dividing by time" ↳

$\underline{p} = (\rho \cdot \text{Area} \cdot l) \cdot \underline{v}$

$\approx \frac{\Delta \underline{p}}{\Delta t} = \rho \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot v$

"dividing by time" ↳ $\underline{F} = (\rho \cdot \text{Area} \cdot l)\cdot \underline{v}$

$\approx \frac{\Delta \underline{F}}{\Delta t} = \rho \cdot \text{Area} \cdot \frac{l}{\Delta t} \cdot \underline{v}$, with $\frac{l}{\Delta t} = \underline{v}$

That is why momentum balance has two velocity terms!

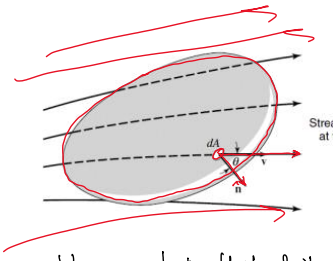
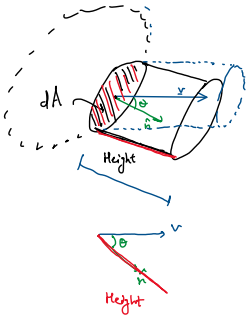


Figure 5.2 Fluid flow through a control volume.

We care about fluid velocity in the direction of the normal to the surface



→ Volume of fluid passed through cross-section A:

$V_{cylinder} = A \cdot \text{Height}$ ← unknown

$v = \frac{\Delta x}{\Delta t}$ → direction of fluid

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$\frac{?}{\Delta x} = \cos\theta$

$? = \Delta x \cos\theta$

$\therefore V_{cylinder} = dA \cdot \Delta x \cdot \cos\theta$

$\Delta V = dA \cdot \Delta x \cos\theta$

$\underline{a}\cdot\underline{b} = |\underline{a}||\underline{b}|\cos\theta$

$\frac{\Delta V}{\Delta t} = dA \cdot \frac{\Delta x}{\Delta t} \cos\theta = A \cdot \underline{v} \cdot \cos\theta$

$\underline{v}$

$\hat{n}\cdot\underline{v} = |\hat{n}||\underline{v}|\cos\theta = \underline{v}\cos\theta$

Volumetric flow rate through A:

$$dQ = v \cdot \cos\theta \, dA = (\hat{n}\cdot\underline{v})\, dA$$

diff. volumetric flow rate

Mass flow = Density · Volumetric flow rate

$dm = \rho \times Q$

$dm = \rho \times (\hat{n}\cdot\underline{v})\, dA$

Momentum flow of liquid through A ⇒ $\underline{F} = (m\underline{v})$

$d\underline{p} = \rho\,(\hat{n}\cdot\underline{v})\, dA\, (\underline{v})$

$(\hat{n}\cdot\underline{v})$: scalar

↳ Now we chose arbitrary area → Taking a differential area & integrating momentum quantity over the entire control surface

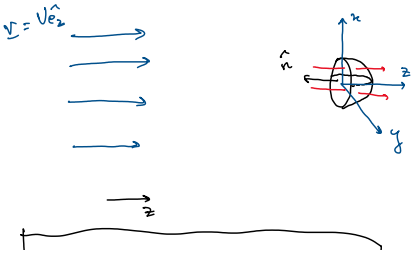
Net momentum change from CV: $\iint_{CS} \underline{v}\,\rho\,(\hat{n}\cdot\underline{v})\, dA$ ⇒ (In − out) = change

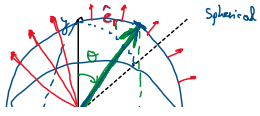
Acc.

control Surface

Problems

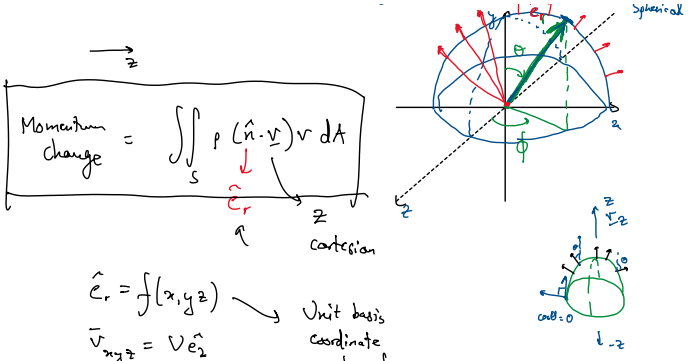
PROBLEM: 3.15 What is the flow of momentum through the hemispherical surface described in Problem 14?





PROBLEM: 3.14 A uniform flow $\underline{v} = U\hat{e}_z$ of an incompressible fluid of density ρ passes through a volume that is in the shape of a half sphere of radius R. The outwards pointing unit normal of the flat surface of the half sphere is $\hat{n} = -\hat{e}_z$. What is the mass flow rate through the hemispherical surface of this volume? Show that you can obtain the correct answer by integrating the formal expression for ∫∫ (Equation 3.87).

R ρ



average velocity

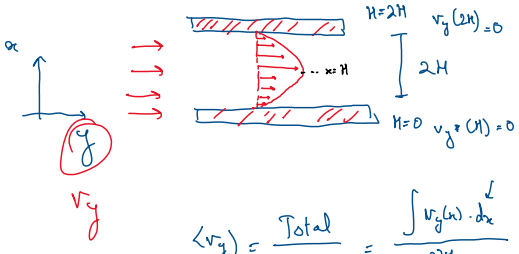
PROBLEM: 3.21 The y-component of a velocity field in flow through a slit (Cartesian coordinates) is given here. What is the average value of the velocity? $2H$ is the gap between the plates. At what location is the velocity a maximum? The units on velocity and A are m/s and the units of x and H are m.

$\underline{v} = \begin{pmatrix}0\\ v_y(x)\\ 0\end{pmatrix}_{xyz}$

$v_y(x) = A\left(1-\frac{(x-H)^2}{H^2}\right)$ — A: m/s; x, H: m

$v_y(x)$ — only in the y direction





$\langle v_y\rangle = \frac{\text{Total}}{\#} = \frac{\int v_y(x)\, dx}{\int_0^{2H} dx} = \frac{\int_0^{2H} v_y(x)\, dx}{2H} = \frac{1}{2H}\int_0^{2H} A\left(1-\frac{(x-H)^2}{H^2}\right) dx$

$\langle v_y\rangle = \frac{2}{3}A$

Maximum velocity

$\frac{\pi}{3}$

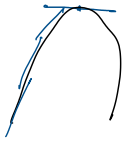
Maximum velocity

$$\frac{dv_y}{dx} = \frac{d}{dx}\left[A\left(1-\frac{(x-H)^2}{H^2}\right)\right] = 0$$

where the max is

$$x = H$$

$$v_y(x=H) = A$$



PROBLEM: 3.31 For the Cartesian coordinate system shown in Figure 3.38, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?

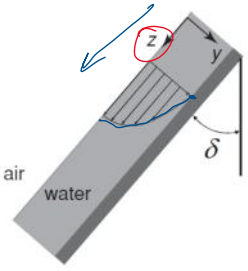
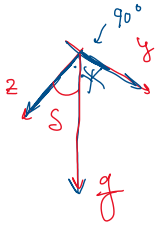


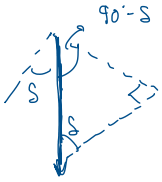
Figure 3.38 Flow coordinate system for Problem 31.



$F = mg$

$$g = \begin{pmatrix} \cdot \\ \cdot \\ g \end{pmatrix}_{xyz}$$

Component in flow direction



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